ARTIFICIAL GRAVITY

LEARNING GOALS

Students will:

- be able to explain how artificial gravity can be created
- solve problems involving artificial gravity

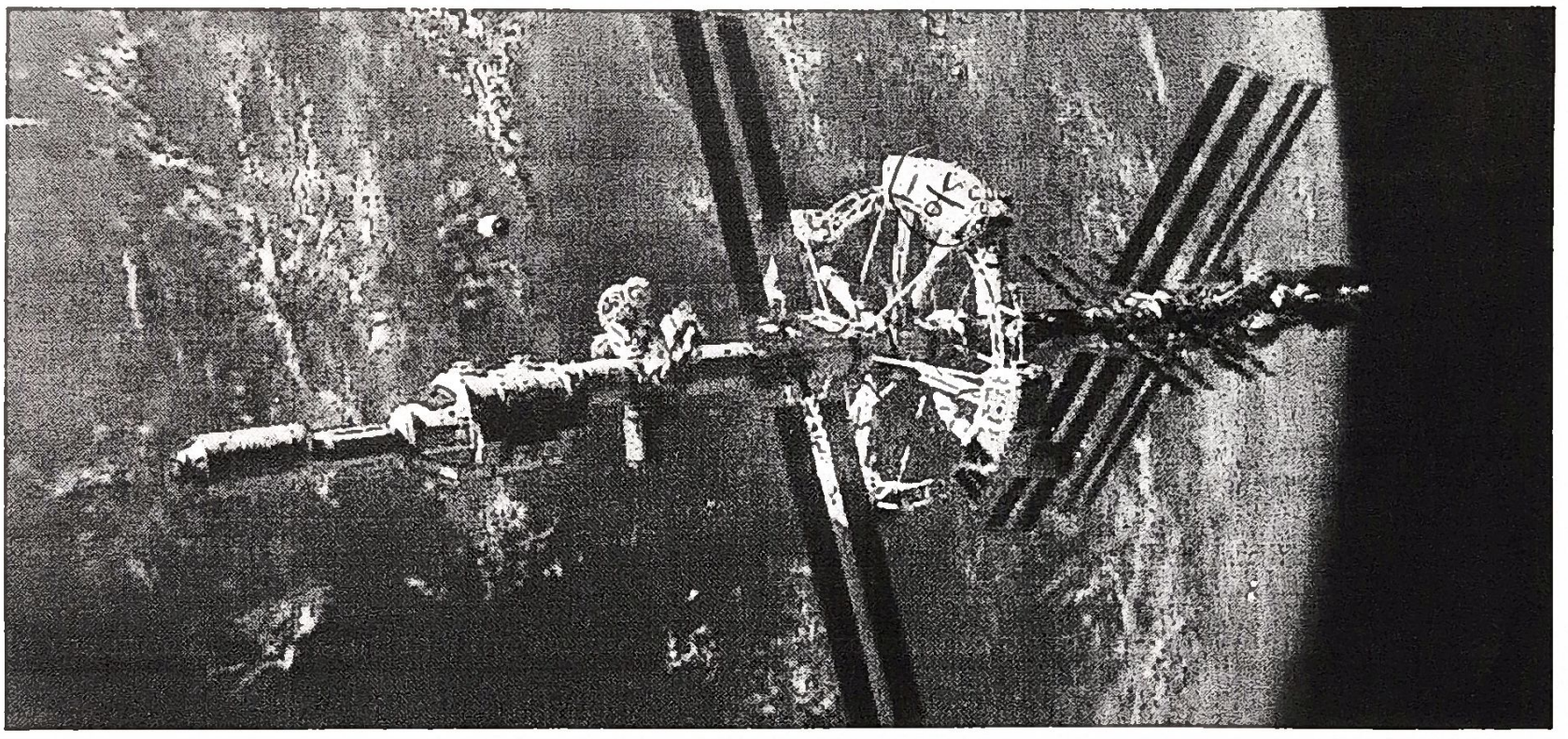
(The Martian 1:36:20; 45:18 left)

APPARENT WEIGHT AND ARTIFICIAL GRAVITY

In order for astronauts to travel interplanetary distances, their spacecraft would need artificial gravity. The effects of constant free fall, or apparent weightlessness, would be disastrous over long periods of time (muscle atrophication, decrease in bone density, fluid build-up in upper body).

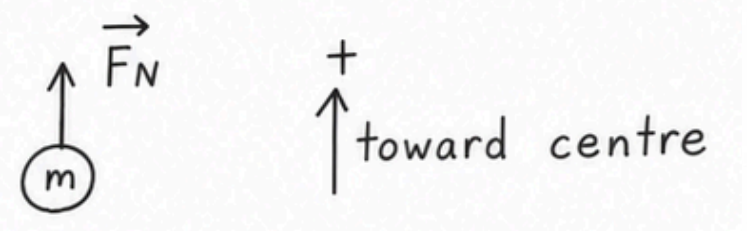
As you know, the sensation of weight, does not come from the force of gravity, rather, it comes from the normal force. One way to produce artificial gravity is to have the spacecraft **rotate**.

In the movie, "The Martian", the *Hermes* spacecraft rotated to produce artificial gravity.



a) Where would an astronaut be standing to get the effects of artificial gravity? Draw this on the picture above. Check with the teacher
Inside of the outer wall of the spinning ring

b) Based on your diagram, draw a free body diagram for the astronaut at this point.



c) In the novel, *The Martian*, the spacecraft creates 0.4g's (i.e. the astronauts experience 0.4 times their regular weight when on the ship.) What does this tell you about their normal force?

Earth: $F_N = mg$

Ship: $F_N = m(0.4g) = 0.4mg$

The normal force is 0.4 times the astronaut's Earth weight.

d) Now estimate the radius of the *Hermes*. Say the pods at the edge of the circle are 3.0 m high.

$r \approx 12\ m$

e) Use your estimate and FBD to calculate the required frequency for the *Hermes* to create the required artificial gravity

$a_c = 4\pi^2 r f^2 \qquad f = \sqrt{\frac{a_c}{4\pi^2 r}}$

$$f = \sqrt{\frac{0.4g}{4\pi^2(12\ m)}} \approx 0.091\ Hz$$

f) If the *Hermes* needed to create 1g of weight (rather than 0.4g) what frequency of rotation would it need? Convert your answer to rpm's (rotations per minute) and discuss how an astronaut might feel on this ship.

$$f = \sqrt{\frac{g}{4\pi^2 r}} \approx 0.144\ Hz$$

$60f \approx 8.6\ rpm$

The floor feels like Earth weight, though rotation may cause dizziness.

g) Now say NASA wanted to reduce materials costs and created a rotating spacecraft with a radius of only 2.0 m. With what frequency would it have to rotate to create 0.4g? What about 1g?

$$f = \sqrt{\frac{0.4g}{4\pi^2 \times 2.0\ m}} \qquad f = \sqrt{\frac{g}{4\pi^2 \times 2.0\ m}}$$

$f \approx 0.223\ Hz$ $\qquad$ $f \approx 0.352\ Hz$

h) Convert your answers in (g) to rpm's and discuss the practicality of this design.

$RPM = 60f$

$RPM \approx 60 \times 0.223$

$RPM \approx 13.4\ rpm$

$RPM = 60f$

$RPM \approx 60 \times 0.352$

$RPM \approx 21.1\ rpm$

The faster spin may cause dizziness, and apparent gravity varies more from head to feet.

ARE ASTRONAUTS ORBITING THE EARTH WEIGHTLESS?

A 57 kg astronaut is on the International Space Station which is orbiting 660 km above the surface of the earth. Weight is DEFINED as the force of gravity acting on an object.

a) Find the weight of the astronaut. [459 N]

$$F_g = \frac{G m_1 m_2}{r^2} = \frac{(6.67\times10^{-11})(5.98\times10^{24})(57)}{(7.04\times10^{6})^2}$$

$$F_g \approx 459\ N$$

b) What is the astronaut's weight on the earth? [559 N]

$F_g = mg \quad F_g = 57\times9.81 \quad F_g = 559\ N$

c) Is the astronaut weightless? If not, what percentage of her earth weight does she have in the space station? [~82%]

No $\frac{458.7}{559.2}\times100 \approx 82\%$

d) Why does the astronaut APPEAR to be weightless even though she clearly has weight?

The astronant falls with the space station, so the floar does not push on her, so $F_N = 0$.

Apparent weight $= F_N$, so the astronaut feels weightless.

CLASS WORK

1. You are an astronaut on a rotating space station. Your station has an inside diameter of 3.0 km
   a. Determine the speed required to create 1g of artificial gravity. Include an FBD. [121 m/s]
   b. Determine the period of rotation of the spacecraft. [78 s]

2. The radius of the Skylab module used by NASA is about 6.6m. If they were to spin the module at 9.7 rpm, what fraction of their regular weight would a person experience? [70%]

1(a)



$F_c = F_N$

$\frac{mv^2}{r} = mg$

$v = \sqrt{rg}$

$v \approx 121\ m/s$

1(b)

$T = ?$

$a_c = 4\pi^2 r f^2$

$f = \sqrt{\frac{a_c}{4\pi^2 r}}$

$f = \sqrt{\frac{v^2}{4\pi^2 r^2}}$

$f \approx 0.01287\ Hz$

$\frac{1}{f} = T$

$T \approx 78\ s$

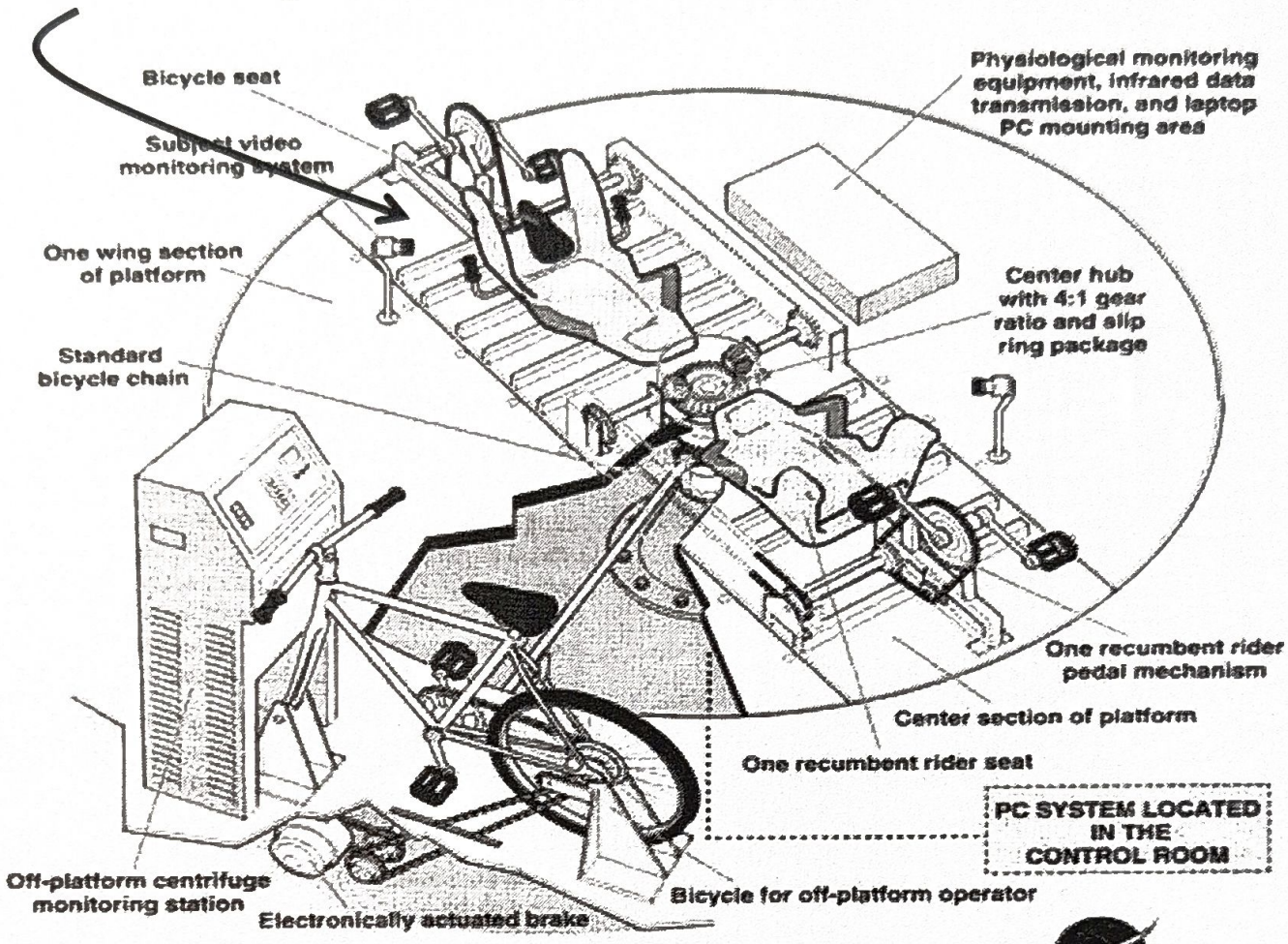
2) $f = 9.7/60\ Hz$

$\frac{F_N}{F_g} = \frac{a_c}{g} = 4\pi^2 r f^2 / g$

$= \frac{4\pi^2(6.6)(9.7/60)^2}{9.81}$

≈ 0.69 (about 70%)

3. In a bicycle centrifuge with a 4.0 m diameter, an astronaut is pedalling while laying down as shown in the diagram below. They create enough force to equal 0.85 g's.



Human Powered Centrifuge

a. Determine the speed, frequency and period of the centrifuge. [4.1 m/s; 0.32 Hz; 3.1 s]
b. Does the speed you calculated in (a) apply to the astronaut's whole body, head or feet? Explain. What about the frequency and period?
c. What might be a drawback to this design?
d. How would the frequency change if the diameter was increased to 6.0 m? Would this make the astronaut feel more dizzy or less dizzy?

4. p 129 #1, 2, 4, 5

a)

$a_c = v^2/r$

$v = \sqrt{a_c r}$

$v = \sqrt{0.85 \times 9.81 \times 2}$

$v = 4.1 \text{ m/s}$

$a_c = \frac{4\pi^2 r}{T^2}$

$T = \sqrt{\frac{4\pi^2 r}{a_c}}$

$T = \sqrt{\frac{4\pi^2 \times 2}{0.85 \times 9.81}}$

$T = 3.1 \text{ s}$

$f = \frac{1}{T}$

$f = \frac{1}{3.1}$

$f = 0.32 \text{ Hz}$

b) The speed is only for the feet because the rest of the body is closer to the centre, so it travels less and moves slower. The same period and frequency apply to the entire body.

c) A drawback is that only the feet feel 0.85 g's.

d) The frequency would decrease, and the astronaut would feel less dizzy.

A Popular Mechanics Article quoted Professor John Page: "The smaller the spacecraft is, the faster it has to rotate," he says, "so if you're going to generate gravity, it's got to be done with a very large spacecraft that spins very slowly. The bigger the disk, the slower you can rotate it." Do a calculation to show that a larger radius decreases the necessary frequency.

$a_c = 4\pi^2 r f^2$

$f = \sqrt{\frac{a_c}{4\pi^2 r}}$

Use $a_c = 1\ m/s^2$ for both.

Vary the radius and look at change in frequency.

$r = 2\ m$

$f = \sqrt{\frac{a_c}{4\pi^2 r}}$

$f = \sqrt{\frac{1}{4\pi^2 \times 2}}$

$f = 0.11\ Hz$

$r = 5\ m$

$f = \sqrt{\frac{a_c}{4\pi^2 r}}$

$f = \sqrt{\frac{1}{4\pi^2 \times 5}}$

$f = 0.07\ Hz$

$0.11 > 0.07$

$\therefore$ Larger radius decreases frequency.